Supplementary Material — The Binary System

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1 The Binary System

Computers process and store information using a system based on only two digits: 0 and 1.

This system is called the **binary system** and is particularly suitable for computers because electronic circuits can easily represent two different states.

1.1 Transistors: ON and OFF

The CPU contains millions or billions of tiny electronic components called **transistors**.

For simplicity, we can think of each transistor as a small switch that can be in one of two states:

- **ON**: current flows through the transistor.
- **OFF**: current does not flow through the transistor.

These two states can be represented using binary digits:

- 1 = ON
- 0 = OFF

A single binary digit is called a **bit**.

2 Positional Number Systems

In a positional number system, the value of a digit depends on its position within the number.

For example, in the decimal number system:

[1] Parts of this document are the result of work carried out by an LLM.

$$153 = 1 \cdot 100 + 5 \cdot 10 + 3 \cdot 1$$

Or, using powers of 10:

$$153 = 1 \cdot 10^2 + 5 \cdot 10^1 + 3 \cdot 10^0$$

The same principle applies to all positional number systems.

3 Binary as a Positional Number System

The binary system is a positional number system with base 2.

This means that each position represents a power of 2.

For example, an 8-bit binary number has the following weights:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

Each bit determines whether its corresponding weight contributes to the total value:

- If the bit is 1, its weight is added.
- If the bit is 0, its weight is not added.

4 Converting Binary to Decimal

To convert a binary number into decimal, we add the weights of all positions containing a 1.

In general, a binary number with bits $b_{n-1}, \ldots, b_1, b_0$ can be converted using the formula:

$$\sum_{i=0}^{n-1} b_i \cdot 2^i$$

where each b_i can be either 0 or 1.

4.1 Example

Let's convert the binary number 11011_2 into decimal.

We add the powers of 2 corresponding to the positions containing a 1:

$$11011_2 = 2^4 + 2^3 + 2^1 + 2^0$$

$$= 16 + 8 + 2 + 1$$

$$= 27_{10}$$

Therefore:

$$\boxed{11011_2 = 27_{10}}$$

5 Converting Decimal to Binary

To convert a decimal number into binary, we need to find the powers of 2 that add up to the original number.

Let's consider the number 153.

We can write it as:

$$153 = 128 + 16 + 8 + 1$$

Using powers of 2:

$$153 = 2^7 + 2^4 + 2^3 + 2^0$$

This means that the bits in positions 7, 4, 3, and 0 must be set to 1.

The remaining positions are set to 0.

Therefore:

$$153_{10} = 10011001_2$$

We can verify the result by converting the binary number back into decimal:

$$10011001_2 = 128 + 16 + 8 + 1 = 153_{10}$$

6 Summary

- The binary system has base 2 and uses only the digits 0 and 1.
- Each bit represents a power of 2.
- To convert binary to decimal, we add the weights of the positions containing a 1.
- To convert decimal to binary, we find the powers of 2 that add up to the original number.

We can think of each power of 2 as a switch that can be either ON (1) or OFF (0).